EFTL: Complex Convolutional Networks With Electromagnetic Feature Transfer Learning for SAR Target Recognition

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***Abstract*— Considering that synthetic aperture radar (SAR) images obtained directly after signal processing are in the form of complex matrices, we propose a complex convolutional network for SAR target recognition. In this article, we give a brief introduction to complex convolutional networks and compare them with the real counterpart. A complex activation function is applied to analyze the influence of phase information in complex neural networks. Inspired by the theory of network visualization, a special kind of transfer learning based on the electromagnetic property from the attributed scattering center model is applied in our networks to modulate the first convolutional layer. The experiment shows a better performance in terms of classification accuracy compared to random weight initialization.**

***Index Terms*— Attributed scattering center model, complex convolutional networks, electromagnetic property, synthetic aperture radar (SAR) target recognition, transfer learning.**

I. Introduction

DEEP learning algorithms are widely used in synthetic aperture radar (SAR) target detection and recognition. Li *et al.* [1] propose a modulated Faster region-convolutional neural networks (R-CNNs) for ship detection. Other applications of deep learning include SAR-optical image matching [2], data fusion [3], and network visualization for SAR target recognition [4]. Wang *et al.* [5] apply all-convolutional networks to SAR data to solve the problems of overfitting in normal convolutional networks. Long short-term memory (LSTM) networks are applied [6] in SAR target recognition to make full use of space-varying scatter information. Zhu *et al.* [7] decompose the raw SAR image data into three components and apply 3-D convolutional neural networks for SAR target classification. Constrained naive generative adversarial networks (CN-GANs) [8] are utilized in target recognition of SAR images to overcome the low signal-to-clutter-noise ratio problem. Some detailed information from SAR image data is applied in deep learning methods, including target rotation [9], pose angle information [10], and speckle noise [11]. Yuan *et al.* [12] apply the local gradient ratio pattern histogram to SAR image target recognition, and Dong *et al.* [13] propose a sparse representation method for SAR target recognition.

Most of the work on deep learning for image classification is based on real-valued optical images [14], [15]; recently, the researchers involved in deep neural networks with complex numbers have received increasing attention [16], [17]. Trabelsi *et al.* [18] applied complex batch normalization (BN) in complex networks for image classification on the dataset from the Canadian Institute for Advanced Research (CIFAR-10). The parameters of the networks are complex-valued, while the inputs are real ones. One of the reasons for complex networks not widely promoted in image classification may be that many large image datasets are real-valued. It is difficult to find a well-labeled SAR dataset, which is large enough to train a complex network for image classification. Wilmanski *et al.* [19] proposed complex input neural networks for SAR automatic target recognition. The first convolution layer is complex-valued followed by an activation function whose output is the magnitude of the input feature map. The rest layers are real-valued. Zhang *et al.* [20] proposed a complex neural network in which all the parameters are complex. They found that complex-valued convolutional neural networks outperform real counterparts in polarimetric SAR (POLSAR) classification [21], [22]. They thought that the phase information is effective for POLSAR classification. Wang *et al.* [23] applied complex residual convolutional neural networks to electromagnetic inverse scattering problems and realized high-resolution electromagnetic imaging. However, most of the recent works were mainly focused on designing a complex neural network to adapt to the complex image data, lacking the analysis on the relationship between the phase information and the network structure to find out an efficient approach to utilize complex phase information.

Pan and Yang [24], for the first time, put forward the concept "knowledge transfer" and divided the transfer learning into three types based on their differences in domains and tasks, including inductive transfer learning, unsupervised transfer learning, and transductive transfer learning. Zamir *et al.* [25] analyzed the relationship of 26 visual tasks and provided tools to compute it. It can help reduce the labeled data requirement for transfer learning but keep the same performance.

Differing from optical images, SAR image data are complex-valued, obtained directly after signal processing with

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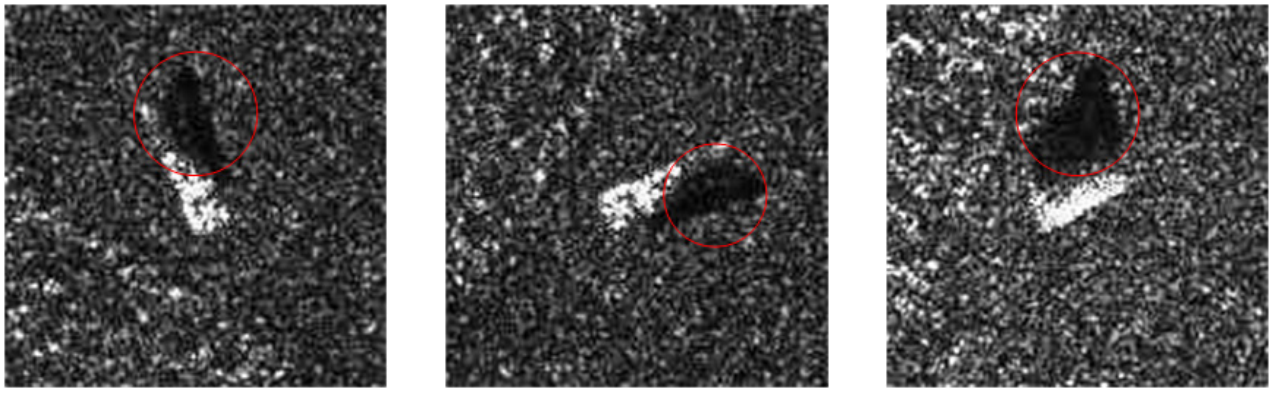
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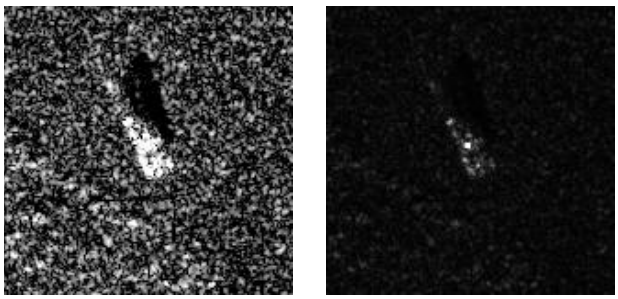
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(a) (b) (c)

Fig. 1. (a) Original image. The direction of the shadow in original image rotating 90° clockwise (b) is different from the same target of which observation angle rotates 90° anticlockwise in reality (c).



(a) (b)

Fig. 2. (a) Logarithmic quantification image has a stronger background but in (b) uniform quantification image; the target is more distinguished.

imaging algorithms, such as the range-Doppler [26], [27], chirp scaling [28], [29], backprojection [30], [31], and so on [32], [33]. Due to these imaging algorithms, SAR images have many special properties that make some common training strategies for optical image datasets not suitable for them, for example, one approach of data argumentation, image rotation (see Fig. 1). Except for the differences resulting from scatter anisotropy [34], [35] in different observation angles, the shadow stays in the same direction while the target rotates [see Fig. 1(c)]. This type of data argumentation aims at making the network learn the rotation feature of the same target. However, if the rotated image [see Fig. 1(c)] is taken into network training and unfortunately, the network learns the feature of the shadow; it will make a wrong decision with higher probability on the truly same target [see Fig. 1(c)] because of the different shadow direction.

Quantification is another problem for the application of real neural networks to SAR target recognition. A gray-scale SAR image is quantified from a floating-point number image. Logarithmic quantification image [see Fig. 2(a)] has a stronger background, while, in uniform quantification image [see Fig. 2(b)], the target is more distinguished. The neural networks might be trained on the dataset of logarithmic quantification, but the images from the recognition task are uniformly quantified. Replacing gray-scale image with the float pointing number image can avoid the problem of quantification. However, without this process, the receiver gain of raw data due to different image sources may lead to other difficulties. The complex neural network with a specific activation function in this article can solve these problems, and we will discuss it in the experiments.

One contribution of this article is the complex activation function. By adjusting the hyperparameter of the complex activation function, we analyze the effect of the phase information in the hidden layers. The other is electromagnetic feature transfer learning (EFTL). Since a complex network has almost double the amount of parameters compared to the real counterpart, it is more difficult to make a complex network convergent from random weight initialization. Transfer learning is a good way to solve this problem. Transfer learning for CNN depends on a large dataset for pretraining, but most of the SAR image datasets are not as large as optical ones. To make matters worse, some typical data argumentation approaches are not appropriate for SAR images as aforementioned. Considering these problems, a special type of transfer learning based on the electromagnetic theory and network visualization is applied in our networks to help modulate the kernels in the first convolutional layer.

The rest of the article is organized as follows. In Section II, we give a brief introduction of complex convolutional neural networks, including the complex convolutional layer, the fully connected layer, the max-pooling layer, activation functions, and softmax classification. We propose an activation function with a hyperparameter to utilize the phase information in the complex neural networks. In Section III, we introduce attributed scattering center model and its further application in EFTL. In Section IV, experiments and discussions are described to analyze the effect of the phase information on network training and testing. Finally, the conclusions are drawn in Section V.

II. Complex Convolutional Neural Networks

A complex number is represented as $z = a + bi$ in the rectangular coordinate system, where a is the real part and b is the imaginary part. a and b here are just used to represent a complex number without any physical significance. On the contrary, a complex number can also be represented as

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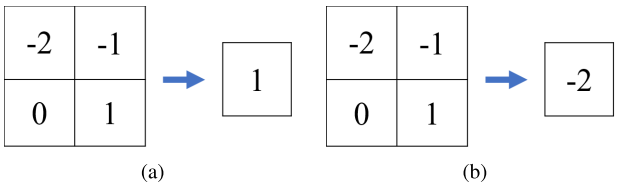


Fig. 3. Difference between the max-pooling layer in a real-valued neural network and the complex one. The maximum in (a) is 1, so the output is 1. In (b), the maximum is 1, but 1 represents $1+0i$. The magnitude of $1+0i$ is 1. -2 represents $-2+0i$, and the magnitude is 2, which is larger than 1, so the output in (b) is -2. The magnitude of the complex number is compared in the max-pooling layer.

$z = |z|e^{i\theta_z}$ in the polar coordinate system, where $|z|$ represents the magnitude of z and θ_z represents the phase. The two formulas of the complex number are equal. However, the form of rectangular coordinate is more convenient in backward propagation for gradient calculation and parameters update. During forward propagation, the form of magnitude and phase in the polar coordinate system is prior for the max-pooling layer and activation functions. In the following content in this section, the two forms are utilized separately to describe different structures in complex-valued neural networks.

A. Forward Propagation

1) Max-Pooling Layer: Convolutional layers and fully connected layers in complex networks are similar to a real network except that the weights are complex. For the max-pooling layer, there is a little difference. In a real-valued max-pooling layer, the output is the maximum among each pooling window. In the complex-valued max-pooling layer, the magnitude is compared instead of the value (see Fig. 3).

2) Complex Activation Function: Numerous activation functions have been proposed for complex neural networks, including modReLU, CReLU, and zReLU [36], [37]. zReLU is defined as follows:

$$\text{zReLU}(z) = \begin{cases} z & \text{if } \theta_z \in [0, \pi/2] \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

θ_z is the phase angle of the input z. The activation function proposed in this article is similar to zReLU, which is defined as $f_{\theta_0}(z)$

$$f_{\theta_0}(z) = \begin{cases} z & \text{if } \theta_z \in [-\theta_0, \theta_0] \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

θ_0 is called the activation angle, which is a threshold to determine whether z should be sent to the next layer or not. As shown in Fig. 4, the solid arrows represent all the variances in the convolutional or fully connected layer, and the imaginary lines represent the activation function. For simplification, convolution can be regarded as another form of affine transformation, which is the same to the fully connected layer

$$y = \sum_{k=1}^{n} (w_k x_k) + b \tag{3}$$

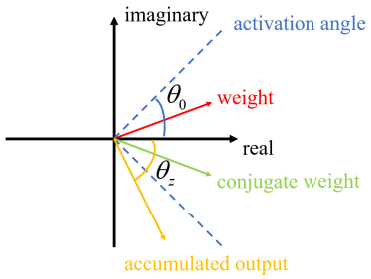


Fig. 4. Complex activation function. The red arrow weight refers to the filters in the convolutional layer or the weight in the fully connected layer. The yellow arrow accumulated output refers to the output feature map from the affine transformation in the convolutional layer or the fully connected layer.

$w_k x_k$ is the input x_k multiplied by the weight w_k. Regardless of the bias b, y is the accumulated output of each $w_k x_k$. If each input has the same phase angle as the conjugate weight, the output along with the accumulated output will appear on the real axis, that is to say, $\theta_z = 0$. Thus, if activation angle θ_0 is small, the accumulated output will be inactive unless the phase angle of each input is close to the conjugated weight so that θ_z can satisfy activation condition $|\theta_z| \leq \theta_0$. In this case, the input that is activated by the function is thought to be similar to the conjugate weight or has a good matching rate with kernel weight.

Besides measuring the similarity between the input feature map and the kernel weight, θ_0 can also adjust the activity ratio, defined as the number of the active feature map dividing the total amount, in a certain layer. For rectified linear unit (ReLU) in a real network, the activity ratio is 0.5 with the hypothesis of zero-mean parameters' initialization. To maintain the activity ratio in a complex network, the activation angle θ_0 can be set to 90°. In fact, 90° is an appropriate value for a complex network to start training but not the best one. The influences of θ_0 are discussed in the experiments.

3) Softmax Regression: Softmax regression is applied in the output layer of deep neural networks to predict the probability distribution over different classes. Hence, we extend the application of softmax regression to complex networks. Softmax regression in a real network is defined as

$$h_{w_i}(x) = \frac{e^{w_i^T x}}{\sum_{i=1}^{k} e^{w_i^T x}} \quad i = 1, 2, \ldots, k. \tag{4}$$

k is the number of classes. x is the input of the softmax classifier. If vector x matches the classification weight w_i or, in the aspect of complex number, is similar to the conjugate weight $\tilde{w}_i$, the input x will more likely belong to the category that w_i represents. However, it is not an easy problem to give a clear definition on similarity. Considering the solution to measure the similarity in complex activation function, $w_i^T x$ in (4) can be replaced with $|\phi_i|$, the absolute phase angle of $w_i^T x$. This is a simple way, but there are two problems. One is that, when $|\phi_i| = 0$, the input vector x point to the same direction as $\tilde{w}_i$. It belongs to the category i with the highest probability, but the output $h_{w_i}(x)$ is the smallest. To avoid the misfit between the output value and the probability distribution, $|\phi_i|$ can be changed into $-|\phi_i|$ or $\cos\phi_i$.

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The other problem is neglect of the magnitude. Though the probability distribution satisfies the equation $\sum_{i=1}^{k} h_{w_i}(x) = 1$ with the input $\cos\phi_i$, for each $h_{w_i}(x)$, the maximum it can approximate to is not 1 because the input is bounded. Besides, if there is another vector X with a larger magnitude, they will share equal probability of the category i resulting from the same phase angle. In fact, we hope X achieve a higher probability. Combining the phase angle and magnitude, alternative input for $w_i^T x$ in complex softmax classification is $|w_i^T x|\cos\phi_i$. Considering that $|w_i^T x|\cos\phi_i = \text{real}(w_i^T x)$, which is the real part of $w_i^T x$, and the imaginary part information denoted by $\text{imag}(w_i^T x)$ has the same problem of the misfit, so softmax regression for complex neural networks is defined as

$$h_{w_i}(x) = \frac{e^{\text{real}\left(w_i^T x\right)}}{\sum_{i=1}^{k} e^{\text{real}\left(w_i^T x\right)}} \quad i = 1, 2, \ldots, k \tag{5}$$

$\text{real}(w_i^T x)$ is also the classification result of the complex neural networks and the position of the maximum determine which category the input x belongs to.

B. Backpropagation

In this part, we mainly focus on the differences between complex backpropagation and real one. The gradient of a real-valued loss function with complex-valued inputs or parameters is defined as follows:

$$\frac{\partial L_w(x)}{\partial x} = \frac{\partial L_w(x)}{\partial\, \text{real}(x)} + j\frac{\partial L_w(x)}{\partial\, \text{imag}(x)}. \tag{6}$$

$L_w(x)$ is a real-valued loss function. x is the input, and w represents the parameter. $\text{real}(x)$ and $\text{imag}(x)$ are real and imaginary parts of x denoted by x_r and x_j for simplification, and j is the imaginary unit. (6) is just defined for gradient calculation or parameter update because it is convenient to process the real and imaginary parts independently in programming realization.

Given a complex input $x = x_r + jx_j$ and a complex weight $w = w_r + jw_j$, affine transformation without bias is

$$\begin{aligned} y = wx &= \left(w_r + jw_j\right)\left(x_r + jx_j\right) \\ &= \left(w_r x_r - w_j x_j\right) + j\left(w_r x_j + w_j x_r\right) = y_r + jy_j. \end{aligned} \tag{7}$$

It is assumed that the gradient of the loss function L is

$$\frac{\partial L}{\partial y} = \frac{\partial L}{\partial y_r} + j\frac{\partial L}{\partial y_j}. \tag{8}$$

Since L is a real-valued loss function, $(\partial L/\partial y_r)$ and $(\partial L/\partial y_j)$ are also real numbers. Applying the chain rule to (8), the weight gradient is

$$\begin{aligned} \frac{\partial L}{\partial w} &= \frac{\partial L}{\partial w_r} + j\frac{\partial L}{\partial w_j} \\ &= \left(\frac{\partial L}{\partial y_r}\cdot x_r + \frac{\partial L}{\partial y_j}\cdot x_j\right) \\ &\quad + j\left(-\frac{\partial L}{\partial y_r}\cdot x_j + \frac{\partial L}{\partial y_j}\cdot x_r\right) = \frac{\partial L}{\partial y}\cdot \tilde{x}. \end{aligned} \tag{9}$$

The backpropagated gradient is

$$\begin{aligned} \frac{\partial L}{\partial x} &= \frac{\partial L}{\partial x_r} + j\frac{\partial L}{\partial x_j} \\ &= \left(\frac{\partial L}{\partial y_r}\cdot w_r + \frac{\partial L}{\partial y_j}\cdot w_j\right) \\ &\quad + j\left(-\frac{\partial L}{\partial y_r}\cdot w_j + \frac{\partial L}{\partial y_j}\cdot w_r\right) = \frac{\partial L}{\partial y}\cdot \tilde{w}. \end{aligned} \tag{10}$$

Compared with the weight gradient in real-valued backpropagation $(\partial L/\partial w) = (\partial L/\partial y)\cdot x$ and $(\partial L/\partial x) = (\partial L/\partial y)\cdot w$, the only difference is the conjugate of x and w. This can help obtain a complex backpropagation formula from a real one. Taking softmax regression as an example, the cross-entropy loss function $l_w(x)$ for real softmax regression and its weight gradient are

$$l_w(x) = -\frac{1}{m}\left[\sum_{i=1}^{m}\sum_{j=1}^{k} 1\left\{y^{(i)} = j\right\}\log\frac{e^{w_j^T x^{(i)}}}{\sum_{l=1}^{k} e^{w_l^T x^{(i)}}}\right] \tag{11}$$

$$\begin{aligned} \frac{\partial l_w(x)}{\partial w} &= -\frac{1}{m}\sum_{i=1}^{m}\left(\frac{\partial l_w(x)}{\partial\left(w^T x^{(i)}\right)}\cdot x^{(i)}\right) \\ \frac{\partial l_w(x)}{\partial\left(w^T x^{(i)}\right)} &= 1\left\{y^{(i)} = j\right\} - \frac{e^{w_j^T x^{(i)}}}{\sum_{l=1}^{k} e^{w_l^T x^{(i)}}}. \end{aligned} \tag{12}$$

In complex softmax regression, $w^T x$ is replaced by $\text{real}(w^T x)$, and the loss function $L_w(x)$ is $-(1/m)\sum_{i=1}^{m}\sum_{j=1}^{k} 1\{y^{(i)} = j\}\log(e^{\text{real}(w_j^T x^{(i)})}/\sum_{l=1}^{k} e^{\text{real}(w_l^T x^{(i)})})$.

Considering the difference between the real backpropagation and the complex one in affine transformation in (9), in complex backpropagation, (12) should be changed by

$$\frac{\partial L_w(x)}{\partial w} = -\frac{1}{m}\sum_{i=1}^{m}\left(\frac{\partial L_w(x)}{\partial\left(w^T x^{(i)}\right)}\cdot \tilde{x}^{(i)}\right). \tag{13}$$

With the definition (6) of the complex value gradient, (13) is expanded as $(\partial L_w(x)/\partial(w^T x^{(i)}))$=$(\partial L_w(x)/\partial(\text{real}(w^T x^{(i)})))$ $+j(\partial L_w(x)/\partial(\text{imag}(w^T x^{(i)})))$. Noting that only the real part of $w^T x$ appears in the loss $L_w(x)$,so $(\partial L_w(x)/\partial(\text{imag}(w^T x^{(i)}))) = 0$. Referring to (12), the weight gradient of complex softmax regression is $(\partial L_w(x)/\partial w) = -(1/m)\sum_{i=1}^{m}[(1\{y^{(i)} = j\} - (e^{real(w_j^T x^{(i)})}/\sum_{l=1}^{k} e^{real(w_j^T x^{(i)})}))$ $\cdot\tilde{x}^{(i)}]$.

If the program environment supports complex value, the real backpropagation can be changed into complex one by replacing the parameters with complex values and alter the input x with $\tilde{x}$ when calculating the weight gradient. The backpropagation of affine transformation, which is common between the adjacent layers, can be derived more conveniently with (9) and (10) from their similar formulas in real networks. Others can refer to the complex chain rule concluded by [1], which is divided into real and imaginary parts. A complex variable $y = y_r + jy_j$ is the function of a complex value $x = x_r + jx_j$.

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If $f(y)$ is a real-valued function, considering (6), we have

$$\begin{aligned}\frac{\partial f(y)}{\partial x} &= \frac{\partial f(y)}{\partial x_r} + j\frac{\partial f(y)}{\partial x_j} \\ &= \frac{\partial f(y)}{\partial y_r}\cdot\frac{\partial y_r}{\partial x_r} + \frac{\partial f(y)}{\partial y_j}\cdot\frac{\partial y_j}{\partial x_r} \\ &\quad + j\left(\frac{\partial f(y)}{\partial y_r}\cdot\frac{\partial y_r}{\partial x_j} + \frac{\partial f(y)}{\partial y_j}\cdot\frac{\partial y_j}{\partial x_j}\right). \end{aligned} \tag{14}$$

Equation (14) is called the complex chain rule.

III. EFTL Based on Attributed Scattering Center Model

One method of transfer learning is to pretrain the CNN on a large assistant dataset and fine-tune the model to a specific task [38]. It is an approach to optimize weight initialization. However, SAR image datasets are usually not so large as optical ones. For example, the Moving and Stationary Target Acquisition and Recognition (MSTAR) dataset has no more than 3000 samples for ten kinds of armored vehicle targets under a certain elevation angle. Compared to the tenth of thousands of images in an optical dataset, MSTAR is not enough to pretrain a deep neural network.

This type of transfer learning aims at enhancing the ability to extract features from training datasets and achieving better performance on another task that is relevant [39]. If the features extracted by CNN can be confirmed, it may be possible to initialize the parameters with other theories, not just restricted to pretraining on a dataset. Deconvolutional networks, as proposed by Zeiler *et al.* [40], project the feature activations from a certain input image back to the input space to study what neural networks have learned in the hidden layers. According to the deconvnet results, the first convolutional layer shows some linear structure and edge features of the images (see Fig. 5). Other relevant network visualization algorithms include class activation mapping (CAM) [41], gradient-weight CAM [42], and so on. Luan *et al.* [43] find that some of the kernels in the first convolutional layer in Alexnet are similar to Gabor filters. They utilize the Gabor filter in the convolutional neural network to reduce the learned parameters and make the learned features robust. They call the network Gabor convolutional networks (GCNs). The 2-D Gabor filter is defined by a plane wave multiplied by a Gaussian function

$$\begin{aligned} G(x,y) &= \exp\left(-\frac{x_0^2+\gamma^2 y_0^2}{2\sigma^2}\right)\times\exp\left(j\frac{2\pi x_0}{\lambda}\right) \\ x_0 &= x\cos\theta + y\sin\theta \\ y_0 &= x\sin\theta + y\cos\theta. \end{aligned} \tag{15}$$

x and y represent the pixels in the space domain of an image, θ represents the orientation of the stripes, σ is the standard deviation of the Gaussian envelope, λ represents the wavelength of the plane wave, and γ is the spatial aspect ratio.

Inspired by the network visualization theory and Gabor filters, EFTL is an application of transfer learning by modulating the filters in the first convolutional layer to extract linear structure features from SAR images based on the attributed scattering center model from the electromagnetic theory.

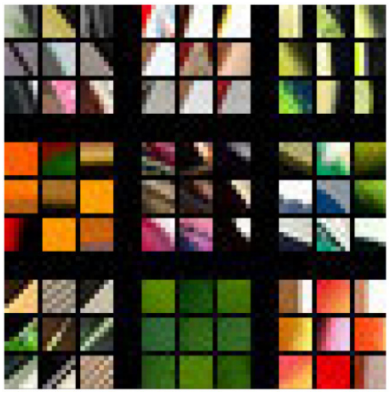


Fig. 5. Filters in the first convolutional layer in deconvolutional networks show some linear structure and edge features of the input images.

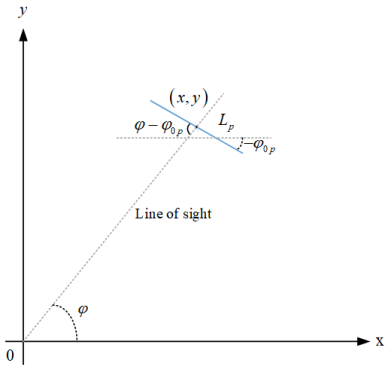


Fig. 6. Attributed scattering center model.

The attributed scattering center model can be approximated by [44]

$$\begin{aligned} S_p(f,\phi) &\approx A_p\,\mathrm{sinc}\left[2\frac{f}{c}L_p\sin(\phi-\phi_{0p})\right] \\ &\quad\times\exp\left[-j4\pi\frac{f}{c}(x_p\cos\phi + y_p\sin\phi)\right]. \end{aligned} \tag{16}$$

$\mathrm{sinc}(x) = (\sin(\pi x)/(\pi x))$, p represents the scattering center; f is the frequency of the signal measured in Hz; ϕ is cross-range observation angle; A_p is the backscatter; c is speed of light; and L_p and ϕ_{0p} are the length and the pointing angle that can be considered as the linear structure feature of the scattering center p (see Fig. 6).The differences between the Gabor filter and (16) is replacing the gaussian function with the sinc function in the frequency domain to extract orientation and scale features from SAR images.

The raw data of SAR are modeled by

$$E(f,\phi) = \sum_p S_p(f,\phi) + n(f,\phi). \tag{17}$$

$n(f,\phi)$ denotes the clutter and noise. Equation (17) can be converted to an image with the R-D imaging algorithm by applying the fast Fourier transform (FFT). Thus, the SAR image is the superposition of several scatters in the spatial domain.

Since Gabor filters in the convolutional neural networks can extract linear structure and edge features from optical images,

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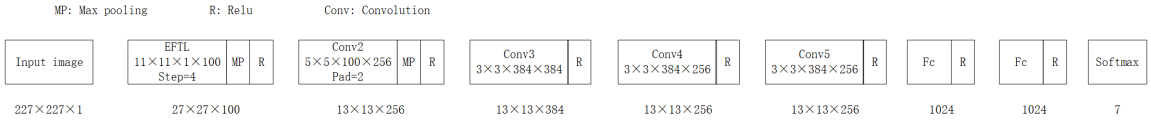


Fig. 7. Neural network that we use in the experiment is similar to Alexnet.

we think that the filters in the complex-valued convolutional neural networks can extract orientation and scale features, which is similar to linear structure in optical images, from SAR images. We utilize attributed scattering center model (16) to modulate the filters in the first convolutional layer; we put the scatter in the center of the image, which means that x_p and y_p are set to zero. We normalize the model with $A_p = 1$ to keep magnitude of all kernels equal. The modified model is defined as

$$\tilde{S}_p(f,\phi) = \operatorname{sinc}\left[2\frac{f}{c}L_p \sin\left(\phi - \phi_{0p}\right)\right]. \tag{18}$$

$\{f, \phi, L_p, \phi_{0p}\}$ remain to be set. f in (18) equals to the same parameter of the training dataset. The sampling frequency is also required to determine the number of the zeros padded for f. $\phi = [-\Delta\phi/2, \Delta\phi/2]$, where $\Delta\phi$ is the variance observation angle of the data. These parameters must be available from the training dataset if applying EFTL to kernel weight initialization. L_p depends on the resolution of the image and the row or column size of the kernel. The resolution refers to the distance in reality that the adjacent two pixels in the image represent. It is assumed that the resolution is m and the kernel is $r \times r$; then

$$L_p = pm \quad p = 1, 2, \ldots, r-1. \tag{19}$$

ϕ_{0p} is the feature of pointing angle of the line structure that we hope to extract. The maximum of ϕ_{0p} is 90°, while the minimum is −90°. The training samples in the MSTAR dataset cover all observation angles of the target. Due to the symmetry, ϕ_{0p} is set to 0°~90°. The interval is set to 10°, so $\phi_{0p} = 0°, 10°, \ldots, 90°$. For each combination of $\{L_p, \phi_{0p}\}$, we apply zero-padding to both sides of f in (19) until the fundamental frequency of f equals to the sampling frequency

$$S(f,\phi) = \operatorname{rect}\left(\frac{f - f_c}{B}\right)\tilde{S}_p(f,\phi). \tag{20}$$

f_c is the carrier frequency, and B is the bandwidth. rect() is the gate function

$$\operatorname{rect}(x) = \begin{cases} 1 & |x| \le 0.5 \\ 0 & |x| > 0.5 \end{cases}. \tag{21}$$

The 2-D-FFT is applied to $S(f,\phi)$, denoted by FS_p. FS_p has an equal size to the input image because the sampling frequency and ϕ are the same as the training dataset. We crop the center of FS_p with the same number of rows or columns of the kernel in the first convolutional layer for weight initialization. The steps for EFTL are given as follows.

1) Obtain f and ϕ from the training dataset.
2) Set up L_p and ϕ_{0p} based on the image resolution and the convolutional kernels.
3) Apply zero-padding to f in (18) to satisfy the sampling frequency of the dataset.
4) Apply 2-D-FFT to (20) and crop the center of FS_p according to the row or column size of convolutional kernels.
5) Initialize the kernel in the first convolutional layer with the cropping patterns.

IV. Experiments

The network in all the experiments in this part has the same framework, which is similar to Alexnet (see Fig. 7) without local response normalization (LRN) and dropout. We choose this typical convolutional neural network because the visualization experiments for the optical dataset are carried out on this neural network. The parameters in convolutional layers and fully connected layers are complex-valued. We use MATLAB to design the structures with small computation capacity, such as the activation function for analyzing phase information and adjusting the activation angle. We refer to Caffe to design complex-valued convolutional layers by replacing the real-valued function with the corresponding one for complex value in Compute Unified Device Architecture (CUDA).

1) EFTL Modulation: The part of the MSTAR dataset applied in this experiment includes seven types of military targets. It contains complex-valued image data and their detailed parameters in the imaging process. The carrier frequency is 9.6 GHz, and the bandwidth is 0.49 GHz, so f in (18) should be set as 9.36–9.85 GHz. The sampling frequency is 0.591 GHz. The sampling point number of f is $227 \times 0.49/0.59 \approx 189$, and the number of zero-padding is 38 (19 on each side). $\Delta\phi$ is about 2.8°, and ϕ varies from −1.4° to 1.4°. The sampling point number of ϕ is 227, the same size to the network input. $L_p = 0.3, 0.6, \ldots, 3$ and $\phi_{0p} = 0°, 10°, \ldots, 90°$. To cover each combination of $\{L_p, \phi_{0p}\}$, we change the size of the kernels in the first convolutional layer to $11 \times 11 \times 1 \times 100$. After applying FFT to (18) and cropping the center of them, the magnitudes of all 100 different cropping patterns are shown in Fig. 9.

For EFTL weight initialization, the filters in the first convolutional layer are initialized by the cropping patterns. The real and imaginary parts of weight parameters in the rest convolutional layers are initialized, respectively, from a zero-mean Gaussian distribution, and the standard deviation is 0.01. The fully connected layers are almost the same initialization except that the standard deviation is 0.005. The activation angle is 90°. The size of a minibatch is 32, and the learning rate is 0.01. The number of the total images that we use is 1917, including 90% for training and the rest 10% for validation. The pitch angle is 15°. The images are padded with zeros around to keep their size equal to

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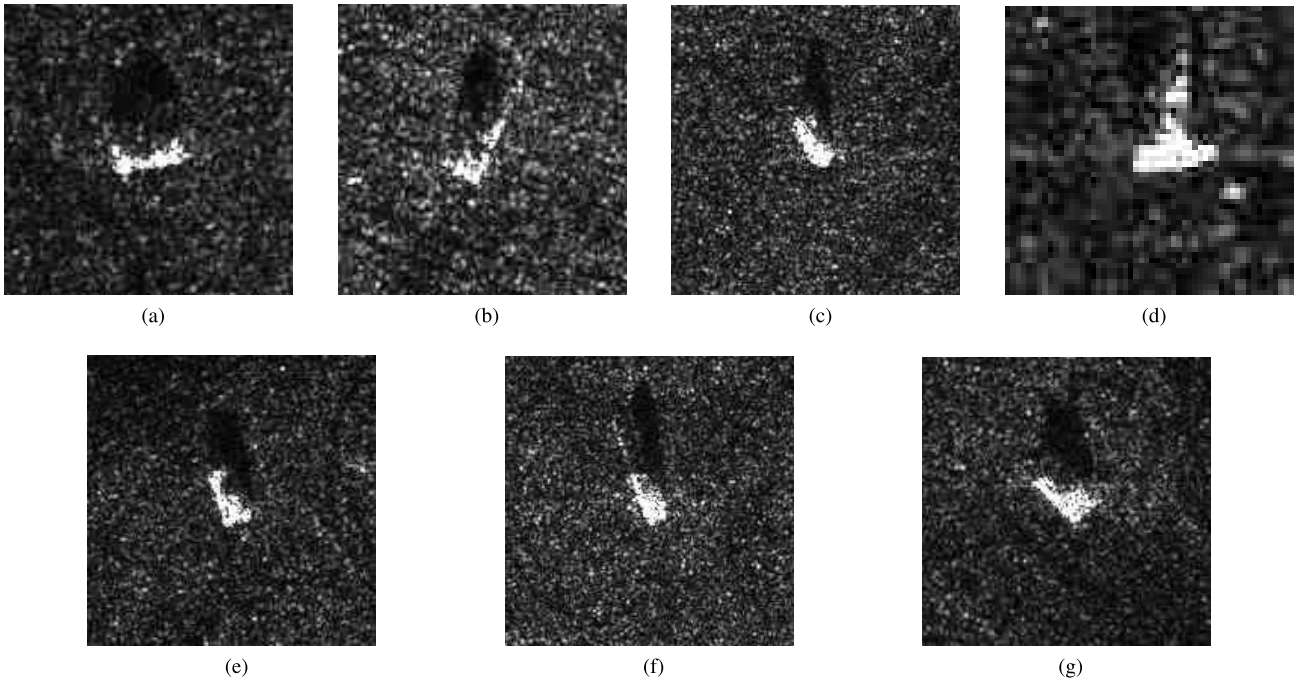


Fig. 8. Dataset for the classification experiment. (a) 2S1. (b) Armored reconnaissance vehicle (BRDM2). (c) D7. (d) Artificial blindage (SLICY). (e) T62. (f) ZIL131. (g) ZSU234.

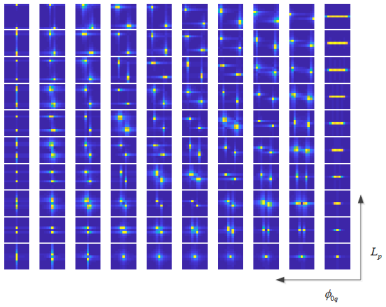


Fig. 9. Amplitude of the filters in the first convolution layer modulated by EFTL with $\phi_{0p} = 0°, 10°, \ldots, 90°$ and $L_p = 0.3, 0.6, \ldots, 3$. Only when $\varphi_{0q} = 0$, the filters contain a straight line (the last column). The line is broken off when φ_{0q} is over zero.

the network input. Only preprocess are dividing the max magnitude of the complex number in each input image. We use the stochastic gradient descent (SGD) algorithm to update the weight in all layers except the first convolutional layer. We stop the training after 1200 iterations (about 20 epochs).

For random initialization and Gabor filters, the difference is the first convolutional layer. In random initialization, the real and imaginary parts of weight parameters are initialized from a zero-mean Gaussian distribution, and the standard deviation is 0.01. In Gabor filters, the filters in the first convolution layer are initialized by (15). The detailed parameters are

$$\begin{aligned} \lambda &= 2, 3, \ldots, 11 \\ \theta &= 0°, 10°, \ldots, 90° \\ \sigma &= 0.56\lambda \\ \gamma &= 0.5. \end{aligned} \tag{22}$$

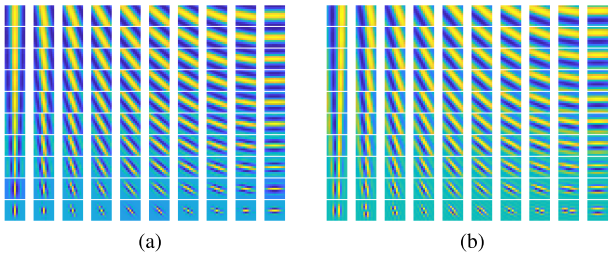


Fig. 10. (a) Real and (b) imaginary parts of Gabor filters.

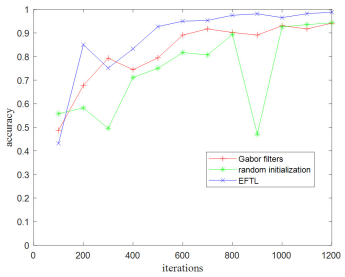


Fig. 11. Classification accuracy on the MSTAR dataset based on three different weight initialization methods, including Gabor filters (red), random initialization (green), and EFTL (blue). Compared with Gabor filters and random initialization, EFTL has a better classification accuracy and converges faster during the same iterations.

We crop the center of the filters with each combination of $\{\lambda, \theta\}$ in size 11×11, so the amount of the filters is 100. The real and imaginary parts of Gabor filters applied in the first convolutional layer are shown in Fig. 10. The result for the complex network with the random initialization, Gabor filters, and EFTL is shown in Table I. The training evolution chart is shown in Fig. 11.

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TABLE I

CLASSIFICATION ACCURACY (%) BY RANDOM INITIALIZATION, GABOR FILTERS, AND EFTL WEIGHT INITIALIZATION AFTER 1200 ITERATIONS. THE PITCH ANGLE OF THE DATA IS 15°

Type	random initialization	Gabor filters	EFTL weight initialization
2S1	92.3	87.2	99.3
BRDM2	97.8	98.9	99.6
D7	95.3	87.2	98.2
SLICY	100	100	100
T62	85.4	93.0	95.2
ZIL131	92.0	96.7	99.3
ZSU234	97.5	96.0	100
Average	94.3	94.2	98.8

TABLE II

TESTING ACCURACY (%) BY RANDOM INITIALIZATION, GABOR FILTERS, AND EFTL WEIGHT INITIALIZATION AFTER 1200 ITERATIONS. THE PITCH ANGLE OF THE DATA IS 17°

Type	random initialization	Gabor filters	EFTL weight initialization
2S1	88.0	89.6	92.0
BRDM2	99.0	97.3	99.3
D7	91.6	96.7	90.6
SLICY	100	100	100
T62	78.3	72.2	88.6
ZIL131	88.4	96.7	96.0
ZSU234	93.4	93.3	96.0
Average	91.8	92.3	95.1

Since random initialization, Gabor filters, and EFTL weight initialization share the same framework, they have a similar capacity for classification tasks on the same dataset. However, the accuracy of the complex network with EFTL is higher than the one with random initialization. One possible reason for this improvement is that EFTL enables the kernels to extract proper features from the inputs at the beginning of the training process but random initialization cannot until the end of it. Thus, EFTL indirectly shortens the time required for parameters to converge and achieves a better performance during the same time of iterations. Another advantage of EFTL is good orthogonality of the kernel weight brought by different combinations of $\{L_p, \phi_{0p}\}$.

The mean accuracy of the network modulated by EFTL is higher than the Gabor filter. Gabor wavelets have a good property to extract the information from both time and frequency domains. They are similar to the response of simple cells in the human vision system. However, SAR images are different from optical ones because the responses are from the microwave instead of the visible light. The linear structures extracted by the first convolutional layer from optical images are consecutive (see Fig. 5). However, in SAR images only when the pointing angle is zero, an intact linear structure can be observed (see Fig. 9). If the scattering center points to other directions, it will cross the range bin, and the linear structures turn to be broken off. This may be the main reason that EFTL outperforms the Gabor filter in SAR target recognition.

The data with the pitch angle of 17° are applied for testing. The amount of the test samples is 2091. The testing accuracy is shown in Table II. EFTL weight initialization can still achieve a better performance than Gabor filters and random initialization when the pitch angle is changed. This demonstrates the robustness of weight EFTL initialization.

To analyze the effect of EFTL initialization on target rotation, we reduce the ratio of training samples from 90% to 70%. Since the samples of each target in the MSTAR dataset cover all 360° observation angles, reducing the ratio of training samples equals to removing part of the observation angles of the target to analyze the robustness of the proposed method to target rotation, such as in Fig. 1. As shown in Fig. 13, the EFTL initialization achieves a better performance without data argumentation, such as image rotation, when part of the observation angles of the target is removed.

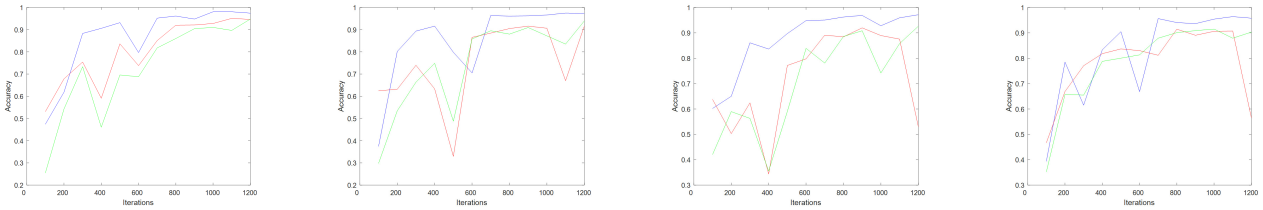
2) Different Activation Angles for Complex Activation Functions: To analyze the effect of the phase information on the classification results, a set of different activation angles for complex activation functions (2), containing $\{90°, 75°, 60°, 45°, 30°\}$, is adopted in the experiments. We perform five independent experiments with random weight initialization on each activation angle. One of the training results after 3600 iterations is shown in Fig. 14. For each activation angle, the mean accuracy of five independent experiments is shown in Fig. 15.

The architecture of the network is the same as Fig. 7, and the weight parameters are randomly initialized. The mean accuracy becomes higher as the activation angle gets smaller from 90° to 60°. The accuracy of the five independent experiments of each activation angle is around their mean value except 45°. In the five experiments of activation angle 45°, the accuracy varies from the minimum 77.31% to the maximum 96.77%. In the activation function, a smaller activation angle leads to more zeros in the output feature map and makes it sparser. The dropout is a typical algorithm making the network sparse to avoid overfitting and help reduce test errors. However, the sparse feature map can also slow down the training. For example, if an activation number of the output feature map is zero, the weight in the following layer connected to this number will have a zero gradient. This is not a serious problem because a weight may be connected with several numbers due to minibatch size or convolution. The gradient of the weight is accumulated by all the numbers connected. However, if there are too many zeros, things may get worse. If all the numbers connected to the weight are zeros, the weight will fail to be trained during this iteration, and more iterations are required. The ratios of the number of zeros are shown in Fig. 16.

In all of the layers in the neural network except the fifth convolutional layer, the zero ratio decreases as the activation angle becomes larger [see Fig. 16(a)]. This proves the hypothesis of the relationship between the activation angle and zero ratios. Though the accuracy changes nearly 20% [see Fig. 16(b)] due to random weight initialization in the four independent experiments, the zero ratio stays the same. Thus, the zero ratios mainly depend on the activation angle.

Considering the relationship between the zero ratios and the activation angle, complex neural networks with this type of activation function have a potential advantage compared to

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(a) (b) (c) (d)

Fig. 12. Classification accuracy in 1200 iterations on the MSTAR dataset based on Gabor filters (red), random initialization (green), and EFTL (blue) with different ratios of training samples, including (a) 85%, (b) 80%, (c) 75%, and (d) 70%. The ratio 90% is shown in Fig. 11.

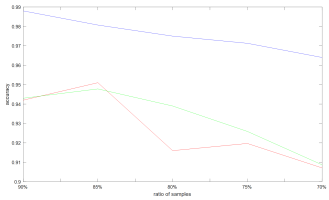


Fig. 13. Classification accuracy [the maximum of each method in Fig. 12, including Gabor filters (red), random initialization (green), and EFTL (blue)] with different ratios of samples.

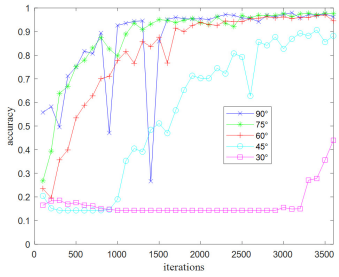


Fig. 14. Training result with five different activation angles after 3600 iterations.

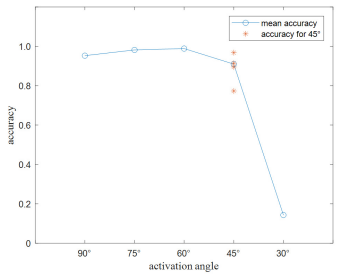
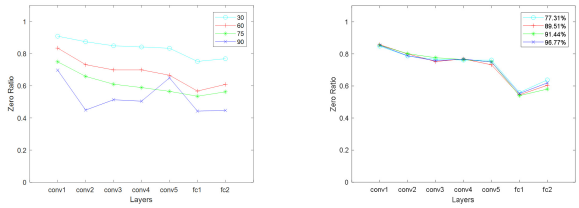


Fig. 15. Classification accuracies of different activation angles. The accuracy reaches the top when the activation angle is 60°, and when it close to 30°, the network has failed to train.

real ones. In real-valued neural networks, the bias b in (3) is initialized with the constant 1 in some layers of the network to accelerate the early periods of learning by providing positive inputs for activation functions. Simonyan *et al.* [45] applied the first-order Taylor expansion to the neural network. It can be proved that, if all the biases are zeros and the activation function is ReLU, the classification result will not be affected by the change of magnitude of the inputs if multiplied by a positive number. During the network training, we may normalize the inputs [46], divided by their maximum



(a) (b)

Fig. 16. Zero ratio of the feature map in different layers with the activation angle (a) $\theta_0 = 30°, 60°, 75°, 90°$. The accuracies changes nearly 20% among four experiments with the same activation angle 45° (b), but the zero ratio in different layers is the same.

and variance or with other rules. In real-valued networks, simply dividing the SAR image by the max magnitude may cause many problems. A_p in (16) satisfies with the equation σ_p=20log$_{10}(4\pi R_0^2 A_p)$. R_0 is the distance between the radar and the center of the target, and σ_p is radar cross section (RCS). RCS represents the scattering properties of the surface, which is an important property in target classification. Thus, we hope to get rid of the effect from R_0 and reserve the RCS. If R_0 in all the input images is the same (including training and testing), since $A_p = 10^{(\sigma_p/20)}/(4\pi R_0^2)$, they can be divided by the max magnitude of all the input images that $A_p/A_{\max} = 10^{(\sigma_p - \sigma_{\max}/20)}$. The maximum should be recorded for testing. However, if R_0 of the input images is different (usually, it is the multisource case), dividing the image by the max magnitude will destroy the information of RCS. Though we can set all the bias to zeros in a real neural network due to the conclusion from first-order Taylor expansion to avoid the effect of R_0, this may cause more feature maps to be inactive and slow down the training. We do not mention the bias in complex neural networks. In fact, the activation function applied in this article can mitigate the bias. Adjusting the activation angle can control the zero ratios of the feature map. At the same time, it can also keep the property that the classification result will not change if the input image is multiplied by a positive number since the bias does not exist truly.

V. Conclusion

Since CNNs have been widely used in target classification tasks for optical images, we extend real-valued convolutional neural networks to the complex domain for SAR target recognition. This study analyzes phase information in complex neural networks and designs the architecture to make full use

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of it. The activation function applied in the complex neural network makes it homogeneous, which is more suitable for complex image data. A special type of transfer learning based on network visualization and electromagnetic scattering model is applied in the complex neural network to enhance the ability of feature extraction from SAR images. The experimental results illustrate the improvement in classification accuracy compared to the random initialization. This provides a new idea for the application of transfer learning in neural networks for SAR target recognition.

REFERENCES

[1] Y. Li, S. Zhang, and W.-Q. Wang, "A lightweight faster R-CNN for ship detection in SAR images," *IEEE Geosci. Remote Sens. Lett.*, early access, Dec. 2, 2020, doi: 10.1109/LGRS.2020.3038901.

[2] L. H. Hughes, N. Merkle, T. Burgmann, S. Auer, and M. Schmitt, "Deep learning for SAR-optical image matching," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2019, pp. 4877–4880.

[3] Y.-R. Cho, S. Shin, S.-H. Yim, K. Kong, H.-W. Cho, and W.-J. Song, "Multistage fusion with dissimilarity regularization for SAR/IR target recognition," *IEEE Access*, vol. 7, pp. 728–740, 2019.

[4] M. Heiligers and A. Huizing, "On the importance of visual explanation and segmentation for SAR ATR using deep learning," in *Proc. IEEE Radar Conf. (RadarConf18)*, Apr. 2018, pp. 0394–0399.

[5] H. Wang, S. Chen, F. Xu, and Y.-Q. Jin, "Application of deep-learning algorithms to MSTAR data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 3743–3745.

[6] F. Zhang, C. Hu, Q. Yin, W. Li, H.-C. Li, and W. Hong, "Multi-aspect-aware bidirectional LSTM networks for synthetic aperture radar target recognition," *IEEE Access*, vol. 5, pp. 26880–26891, 2017.

[7] H. Zhu, T. Wong, N. Lin, W. Wang, and S. Thedoridis, "Synthetic aperture radar target classification based on 3-D convolutional neural network," in *Proc. IEEE 5th Int. Conf. Signal Image Process. (ICSIP)*, Oct. 2020, pp. 440–445.

[8] C. Mao, L. Huang, Y. Xiao, F. He, and Y. Liu, "Target recognition of SAR image based on CN-GAN and CNN in complex environment," *IEEE Access*, vol. 9, pp. 39608–39617, 2021.

[9] S. Zhang, Z. Wen, Z. Liu, and Q. Pan, "Rotation awareness based self-supervised learning for SAR target recognition," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2019, pp. 1378–1381.

[10] J. Oh, G.-Y. Youm, and M. Kim, "SPAM-net: A CNN-based SAR target recognition network with pose angle marginalization learning," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 2, pp. 701–714, Feb. 2021.

[11] Y. Kwak, W.-J. Song, and S.-E. Kim, "Speckle-noise-invariant convolutional neural network for SAR target recognition," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 4, pp. 549–553, Apr. 2019.

[12] X. Yuan, T. Tang, D. Xiang, Y. Li, and Y. Su, "Target recognition in SAR imagery based on local gradient ratio pattern," *Int. J. Remote Sens.*, vol. 35, no. 3, pp. 857–870, Feb. 2014, doi: 10.1080/01431161.2013.873150.

[13] G. Dong, G. Kuang, N. Wang, and W. Wang, "Classification via sparse representation of steerable wavelet frames on Grassmann manifold: Application to target recognition in SAR image," *IEEE Trans. Image Process.*, vol. 26, no. 6, pp. 2892–2904, Jun. 2017.

[14] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2012, pp. 1097–1105.

[15] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[16] J. Gao, B. Deng, Y. Qin, H. Wang, and X. Li, "Enhanced radar imaging using a complex-valued convolutional neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 1, pp. 35–39, Jan. 2019.

[17] Q. Yi, L. Xiao, Y. Zhang, B. Liao, L. Ding, and H. Peng, "Nonlinearly activated complex-valued gradient neural network for complex matrix inversion," in *Proc. 9th Int. Conf. Intell. Control Inf. Process. (ICICIP)*, Nov. 2018, pp. 44–48.

[18] C. Trabelsi *et al.*, "Deep complex networks," *CoRR*, vol. abs/1705.09792, 2017. [Online]. Available: http://arxiv.org/abs/1705.09792

[19] M. Wilmanski, C. Kreucher, and A. Hero, "Complex input convolutional neural networks for wide angle SAR ATR," in *Proc. IEEE Global Conf. Signal Inf. Process. (GlobalSIP)*, Dec. 2016, pp. 1037–1041.

[20] Z. Zhang, H. Wang, F. Xu, and Y.-Q. Jin, "Complex-valued convolutional neural network and its application in polarimetric SAR image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 12, pp. 7177–7188, Dec. 2017.

[21] X. Qin, T. Hu, H. Zou, W. Yu, and P. Wang, "Polsar image classification via complex-valued convolutional neural network combining measured data and artificial features," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2019, pp. 3209–3212.

[22] X. Mei, W. Nie, J. Liu, and K. Huang, "Polsar image crop classification based on deep residual learning network," in *Proc. 7th Int. Conf. Agro-Geoinformatics (Agro-Geoinformatics)*, Aug. 2018, pp. 1–6.

[23] L. G. Wang, M. Wang, W. Zhong, and L. Li, "Complex-valued deep convolutional networks for nonlinear electromagnetic inverse scattering," in *Proc. IEEE Int. Conf. Comput. Electromagn. (ICCEM)*, Mar. 2018, pp. 1–2.

[24] S. J. Pan and Q. Yang, "A survey on transfer learning," *IEEE Trans. Knowl. Data Eng.*, vol. 22, no. 10, pp. 1345–1359, Oct. 2010.

[25] A. Zamir, A. Sax, W. Shen, L. Guibas, J. Malik, and S. Savarese, "Taskonomy: Disentangling task transfer learning," in *Proc. 28th Int. Joint Conf. Artif. Intell.*, Aug. 2019, pp. 3712–3722.

[26] W. Fan, M. Zhang, J. Li, and P. Wei, "Modified range-Doppler algorithm for high squint SAR echo processing," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 3, pp. 422–426, Mar. 2019.

[27] Z. Wang, Y. Li, S. Shao, S. Li, J. Xiao, and J. Wu, "Improved range Doppler algorithm based on squint FMCW-SAR," in *Proc. 12th Int. Conf. Signal Process. Commun. Syst. (ICSPCS)*, Dec. 2018, pp. 1–5.

[28] Z. Zhang, M. Xing, L. Li, J. Zhen, and Z. Bao, "Chirp scaling algorithm for parallel bistatic SAR data processing," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, vol. 2, 2009, pp. II-53–II-56.

[29] Y. Yuan, S. Chen, S. Zhang, and H. Zhao, "A chirp scaling algorithm for forward-looking linear-array SAR with constant acceleration," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 1, pp. 88–91, Jan. 2018.

[30] L. Zhang, H.-L. Li, Z.-J. Qiao, M.-D. Xing, and Z. Bao, "Integrating autofocus techniques with fast factorized back-projection for high-resolution spotlight SAR imaging," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 6, pp. 1394–1398, Nov. 2013.

[31] W. Fan, M. Zhang, and J. Li, "SAR imagery simulation from complicated targets based on back projection algorithm," in *Proc. IEEE Int. Conf. Comput. Electromagn. (ICCEM)*, Mar. 2018, pp. 1–2.

[32] J. Chen, J. Zhang, Y. Jin, H. Yu, B. Liang, and D.-G. Yang, "Real-time processing of spaceborne SAR data with nonlinear trajectory based on variable PRF," *IEEE Trans. Geosci. Remote Sens.*, pp. 1–12, 2021.

[33] J. Chen, M. Xing, X.-G. Xia, J. Zhang, B. Liang, and D.-G. Yang, "SVD-based ambiguity function analysis for nonlinear trajectory SAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3072–3087, Apr. 2021.

[34] Y. Gao and M. Xing, "A method for extracting amplitude attribute of scattering centers in SAR," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 2665–2668.

[35] Y. Li, Q. Yin, Y. Lin, and W. Hong, "Anisotropy scattering detection from multiaspect signatures of circular polarimetric SAR," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 10, pp. 1575–1579, Oct. 2018.

[36] A. Hirose and S. Yoshida, "Generalization characteristics of complex-valued feedforward neural networks in relation to signal coherence," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 23, no. 4, pp. 541–551, Apr. 2012.

[37] S. Scardapane, S. Van Vaerenbergh, A. Hussain, and A. Uncini, "Complex-valued neural networks with nonparametric activation functions," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 4, no. 2, pp. 140–150, Apr. 2020.

[38] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 580–587.

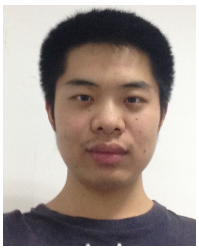
[39] M. Oquab, L. Bottou, I. Laptev, and J. Sivic, "Learning and transferring mid-level image representations using convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 1717–1724.

[40] M. D. Zeiler, G. W. Taylor, and R. Fergus, "Adaptive deconvolutional networks for mid and high level feature learning," in *Proc. Int. Conf. Comput. Vis.*, Nov. 2011, pp. 2018–2025.

[41] B. Zhou, A. Khosla, A. Lapedriza, A. Oliva, and A. Torralba, "Learning deep features for discriminative localization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2921–2929.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 08:44:24 UTC from IEEE Xplore. Restrictions apply.

[42] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 618–626.
[43] S. Luan, C. Chen, B. Zhang, J. Han, and J. Liu, "Gabor convolutional networks," *IEEE Trans. Image Process.*, vol. 27, no. 9, pp. 4357–4366, Sep. 2018.
[44] J. Duan, L. Zhang, M. Xing, Y. F. Wu, and M. Wu, "Polarimetric target decomposition based on attributed scattering center model for synthetic aperture radar targets," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 12, pp. 2095–2099, Dec. 2014.
[45] K. Simonyan, A. Vedaldi, and A. Zisserman, "Deep inside convolutional networks: Visualising image classification models and saliency maps," 2013, *arXiv:1312.6034*. [Online]. Available: https://arxiv.org/abs/1312.6034
[46] V. Thakkar, S. Tewary, and C. Chakraborty, "Batch normalization in convolutional neural networks—A comparative study with CIFAR-10 data," in *Proc. 5th Int. Conf. Emerg. Appl. Inf. Technol. (EAIT)*, Jan. 2018, pp. 1–5.



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